25901121 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOTA00 - APPLIED PHYCOLOGY AND MYCOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **2** marks

1. Name the principal xanthophyll in golden-brown algae. [K] / [CO1]
2. Distinguish between micro and mesophytoplankton [U] / [CO2]
3. Give any two ecological impacts of HAB [U] / [CO3]
4. Does growing algae in large "open ponds" cost less than using high-tech "closed photobioreactors"? [U] / [CO2]
5. Define soxlet extraction? [U] / [CO2]
6. Name any two food items produced from algae [K] / [CO2]
7. What is Agar Agar? [U] / [CO3]
8. Name the major classes of fungal enzymes used in industry. [K] / [CO4]
9. Which mushroom is commonly cultivated for food security? [K] / [CO4]
10. Which organ is primarily affected by afflatoxin exposure. [U] / [CO4]

11. Name on pharmaceutical compound derived from lichen. [U] / [CO4]

12. Give an example of bioactive compounds produced by fungal endophytes. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. List the types of ribosomes and their rRNA composition as used in algal systematics. [K] / [CO1]

14. Expand ATCC and its significance. [K] / [CO2]

15. Describe fungal bio fertilizers with suitable examples. [K] / [CO3]

16. Describe various applications of algal extracts. [A] / [CO2]

17. Explain different molecular genetic techniques for Algal bioengineering. [U] / [CO2]

18. Why is yeast prefered in the baking industry? [U] / [CO3]

19. Explain how fungi serve as model organisms in scientific research. [U] / [CO4]

20. Explain the importance of sanitation in storage management. [U] / [CO3]

21. Explain the concept of neutraceuticals and discuss the role of fungi as neutraceutical source. [U] / [CO4]

22. Explain the concept of bio-prospecting in fungi. [U] / [CO4]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Analyze the future prospects of algal biotechnology. [U] / [CO2]

24. Explain how fungi help in the breakdown of agro-residues. [U] / [CO4]

[10x1 = 10]